

MATH 174 - HW 1

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1. (a) Forward substitution on $Ly = b$

$$\begin{aligned}x_1 &= 2 \\-x_1 + x_2 &= -1 \\3x_1 - 3x_2 + x_3 &= 3\end{aligned}$$

So, we have

$$\begin{aligned}x_1 &= 2 \\x_2 &= -1 + x_1 \\&= -1 + 2 \\&= 1 \\x_3 &= 3 - 3x_1 + 3x_2 \\&= 3 - 3(2) + 3(1) \\&= 0\end{aligned}$$

- (b) Backward substitution on $Ux = y$

$$\begin{aligned}-1x_1 + 2x_2 + x_3 &= y_1 \\2x_2 - 2x_3 &= y_2 \\5x_3 &= y_3\end{aligned}$$

So, we have

$$\begin{aligned}
 x_3 &= \frac{y_3}{5} \\
 2x_2 &= y_2 + 2x_3 \\
 2x_2 &= y_2 + 2\left(\frac{y_3}{5}\right) \\
 x_2 &= \frac{1}{2}y_2 + \frac{y_3}{5} \\
 -x_1 &= y_1 - 2x_2 - x_3 \\
 x_1 &= -y_1 + 2y_2 + \frac{2y_3}{5} + \frac{y_3}{5} \\
 &= -y_1 + 2y_2 + \frac{3y_3}{5}
 \end{aligned}$$

(c) Compute $A = LU$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & -3 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 3 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 2 & 3 \\ 1 & 0 & -5 \\ -3 & 0 & 20 \end{bmatrix}$$

$$\begin{aligned}
 Ax = b : \begin{bmatrix} -1 & 2 & 3 \\ 1 & 0 & -5 \\ -3 & 0 & 20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \\
 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} -1 \\ \frac{1}{2} \\ 0 \end{bmatrix}
 \end{aligned}$$

The solution to $Ax + b$ is any multiple of $\begin{bmatrix} -1 \\ \frac{1}{2} \\ 0 \end{bmatrix}$

2. (a) The number of additions and subtractions needed to perform forward substitution is
- (b)
- (c)
- (d)

(e)

(f)

(g)